

Code-Layout, Readability and Reusability

Date	05 July 2026
Team ID	XXXXXX
Project Name	HDI Predictor (Human Development Index Prediction Using Machine Learning)
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used Throughout the code	Yes	Proper Python indentation followed throughout the project.
2	Proper File Structure	Files and folders are logically organized	Yes	Files organized into templates, static, model, and application files.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables such as life, expected, mean, income, and prediction are meaningful.
4	Function / Method Names	Functions are descriptively named	Yes	Functions like home() and predict() clearly describe their purpose.
5	Code Comments	Inline and block comments explain log c	Partial	Basic comments included; more comments can improve readability.
6	Modular Design	Code is split into reusable functions/module	Yes	Machine learning model and Flask application are separated into different files.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Duplicate or unnecessary code removed.
8	Error Handling	Exceptions and errors are handled gracefully	Partial	Basic input validation exists; advanced exception handling can be added.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Prediction Model	Python, Scikit-learn	Predicting HDI	High

2	Flask Application	Flask	Backend Request Handling	High
3	HTML Template	HTML	User Interface	High
4	CSS Stylesheet	CSS	Page Styling	High
5	Joblib Model Loader	Joblib	Loading Trained Model	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project structure.
Readability	5	Code is simple and easy to understand.
Reusability	5	Components can be reused in future ML projects.
Documentation / Comments	4	Basic documentation provided; additional comments can improve maintainability.
Overall Score	4.8/5	Clean, modular, and maintainable machine learning application.